

Greenroom

The AI copilot for hackathon & community organizers — **with gates.**

Every answer needs a citation. Every outreach needs an approval. Every action leaves a trail.

Built on this hackathon's own published rules. Demo runs on an explicitly labeled synthetic replay.

citation gate

approval gate

audit trail

fail-closed on conflict

THE PROBLEM

3,063 participants. One Discord server.
No operator.

×11

the same rules question, asked again at 2am — organizers answer by hand or not at all

4 days

registered, never joined, never introduced — silent attrition nobody sees until submissions

1 by hand

sponsor engagement report compiled at midnight, numbers nobody can reproduce

Dashboards show the problem but don't act. Chatbots act, but can't be trusted not to invent rules. **Organizers need an operator that is powerful because it is constrained.**

Three deterministic gates. The LLM only writes wording.

gate 1

Citation Gate

Every answer sentence anchors to one official source sentence and stays faithful to it — no flipped "not", no swapped "must", no changed number. Otherwise: deterministic escalation to a human.

gate 2

Approval Gate

The copilot drafts outreach to stalled participants — nothing is ever sent until a human organizer clicks approve. Failed sends retry; crash-interrupted sends are flagged, never silently repeated.

gate 3

Audit Trail

Every answer, refusal, conflict, draft, approval and send is an append-only record. Reports are computed from the trail — numbers you can reproduce by hand.

And when official sources disagree with each other, Greenroom **fails closed**: both sources verbatim, a human route — never a one-sided verdict.

"Can companies participate?"

Greenroom refuses to guess.

official rules prose

"Eligible participants include: ... **Startup founders and entrepreneurs**, Open-source contributors and technology enthusiasts."

`eligibility.rules_text – ai-builders-hackathon-2026.devpost.com/rules`

official structured settings

"**Students only**. Companies/professional organizations **excluded** from participation."

`eligibility.structured – ai-builders-hackathon-2026.devpost.com/rules`

verified by a human

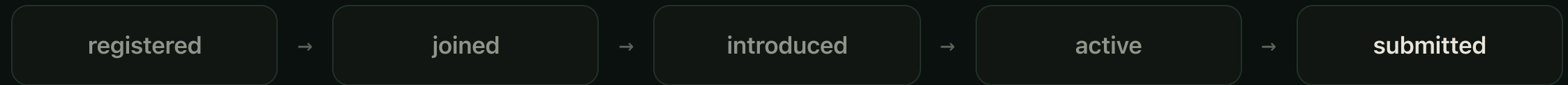
Both sources shown verbatim. Routed to a human organizer. Most bots pick one and answer; an operator admits when the rules disagree.

Deterministic. Per sentence. Not prompt engineering.



- **Merged citations can't smuggle violations** — a flipped "never mandatory" hiding behind a faithful sentence is still caught; each claim sentence is anchored on its own.
- **Stale information never ships as current** — a passed application deadline is acknowledged, not offered.
- **One repair pass with the gate's exact reasons** — wording gets fixed; fabrications still escalate.

Every participant is a state machine. Every nudge waits for a human click.



- **Stall radar** flags who is stuck where — never-joined, never-introduced, gone-quiet, missing-submission — and drafts a nudge matched to the reason.
- **Nothing sends itself** — drafts sit in the approval queue until an organizer approves or rejects with a reason.
- **Honest delivery states** — sent requires a real receipt; the demo sender is labeled `simulated`; failures land in `send_failed` and retry idempotently; a crash mid-send is recovered as "outcome unknown", never silently resent.

Numbers a sponsor can reproduce.

One click, computed from the trail

- Participation funnel & 72h activity
- Copilot support load: answered / escalated / **conflicted** counted separately
- Outreach: drafted, approved, **sent vs simulated never merged**, failed, rejected
- Sponsor deliverables with anchor links and reach

Radical honesty by default

- Every figure recomputes from `audit.jsonl` — open the file, check the math
- Demo data is marked **synthetic replay** on the report itself
- Unverified conflict candidates wait in a review queue
 - they never reach participants

The gates are tested, not promised.

73/73

unit tests — incl. an adversarial polarity suite: not/never flips, must↔may swaps, dropped exclusions, date & number drift, and the same violations merged into a single citation claim

13/13

real-browser E2E — fail-closed money moment, escalation, approval → simulated send, audit, report

23/23

live eval against hosted Kimi K2.7 — paraphrases, typos, non-English, stale traps, fabrication bait; zero over-escalation

- **Generalization** — the identical pipeline, zero tuning, answers a second hackathon's rules (Nebius × NVIDIA) — including the opposite eligibility answer.
- **Evidence is bound to the code** — every artifact records the source-tree hash, corpus hash and tool versions; recompute the hash at HEAD to verify.

All decisions deterministic. Providers swap; permissions don't.

core — pure logic, zero I/O

- BM25 retrieval · per-sentence citation gate
- conflict scanner: detector proposes, human registry disposes
- lifecycle state machine · stall rules · approval queue
- append-only audit · deterministic report

edges

- **LLM** — Kimi K2.7 hosted $\Leftrightarrow$ deterministic offline mock, same gate
- **Server** — Fastify API, JSON/JSONL persistence, atomic snapshots
- **Dashboard** — React organizer workbench
- **Discord adapter** — implemented (tri-state replies, DM receipts); live-guild pilot is next

Privacy is a feature: radar reads only public-channel metadata; outreach is opt-in and human-approved.

Usable by this hackathon's organizers tomorrow — and honest about today.

real today

- Full loop offline: question → cited answer / fail-closed → stall flag → approved nudge → audit → sponsor report
- A genuine, human-verified conflict surfaced in this event's own rules
- Cold-start reproducible: install → test → build → browser E2E

next

- Live-guild Discord pilot (adapter implemented; wiring & verification next)
- Real registration/submission feeds with a proper data contract
- Organizer pilot measuring response time and stall-recovery

github.com/er-s-an/greenroom — README has the 2-minute verification script.

Demo data is synthetic and labeled as such. No adoption or live deployment is claimed.